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A novel ST-iTransformer model for spatio-temporal ambient air pollution forecasting

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Abstract

Accurate prediction of air pollutant dynamics is crucial for public health, sustainable environmental management, and ecosystem analysis. Air pollutants exhibit intricate spatial and temporal patterns, which pose considerable challenges for accurate prediction. This study proposes a novel Spatio-temporal inverted Transformer (ST-iTransformer) model, which uses a new spatio-temporal embedding and inverted dimensions for attention mechanisms to improve air pollution forecast accuracy. The input is three-dimensional data comprising multi-station, multi-step air pollutants, and meteorological variables. In the first phase, the input data is transformed by the proposed spatio-temporal embedding. Next, attention and feed-forward network strategies are applied to extract spatio-temporal features from the embedded data. Then, the multi-station, multi-step air pollutants forecast results of all the monitoring stations are obtained based on a linear layer. Systematic experiments conducted in Beijing, China, show that ST-iTransformer significantly outperforms benchmark models across the metrics, air pollutants, and stations. Additionally, ablation experiments confirm that the spatio-temporal embedding, the inclusion of spatial data, and the addition of meteorological data all improve the prediction accuracy of the model. These results demonstrate that the proposed model could sufficiently capture the spatio-temporal correlation characteristics of air quality data and be successfully applied for air quality forecasting.

Keywords: Air pollution, Spatio-temporal prediction, Transformer, Embedding, Deep learning

Introduction

Outdoor air pollution is an important environmental health issue that affects people in almost all countries. The World Health Organization (WHO) estimated that in 2019, exposure to ambient air pollution in urban and rural areas caused around 4.2 million premature deaths globally each year. Common air pollutants include a range of substances that can have significant impacts on human health and the environment. Particulate matter (PM) is a major concern, with PM_{2.5} and PM₁₀ referring to fine particles with diameters of 2.5 micrometers or less and 10 micrometers or less, respectively. Exposure to fine particulate matter leads to cardiovascular and respiratory diseases, as



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well as cancers [1]. Other prevalent pollutants include carbon monoxide (CO), ozone (O₃), nitrogen dioxide (NO₂), and sulfur dioxide (SO₂). Each of these pollutants poses unique risks to health and the environment, making their monitoring and control crucial for air quality management.

Performing accurate forecasts of air pollutants is an effective way to avoid exposure to pollution [2]. Forecasts enable decision-makers to promptly implement pollution control measures, alert vulnerable populations to protect public health, and develop sustainable urban policies to mitigate long-term environmental damage. The significance of advanced predictive models is evident in their ability to shape proactive and informed strategies for public health protection and sustainable ecological management.

Air pollutants forecasting and warning systems have become prominent research fields in recent decades. Researchers have employed numerous strategies, including knowledge-based and data-driven models. Knowledge-based models include Community Multiscale Air Quality [3], Weather Research and Forecasting Models and Chemistry [4], and the Nested Air Quality Predictive Modeling System [5]. The significant computational resources and specialized expertise limit their practicality for real-time applications [6, 7]. Data-driven models provide more computational efficiency and give notable benefits for predicting future unknown air pollutant values based on existing data. Data-driven models are divided into three categories. The first category includes classical statistical models, such as Autoregressive Integrated Moving Average model [8], Multiple Linear Regression [9], and Geographically Weighted Regression [10]. The second category consists of traditional machine learning models, which include the Artificial Neural Network [11], Support Vector Machine [12], and Random Forest [13]. The third category encompasses deep learning models, such as Gated Recurrent Unit [14], Long Short-Term Memory (LSTM) [15], and Transformer [16]. Deep learning models achieve more accurate forecasting due to their more flexible structures, more robust feature extraction capabilities, and higher non-linear processing abilities [17, 18]. These advantages have led to increased adoption of deep learning approaches in recent research on prediction of air pollutants.

In the field of spatio-temporal air quality prediction, researchers have explored various deep learning architectures, which can be systematically categorized into three main groups: Convolutional Neural Network (CNN)-based composite models, Graph Neural Network (GNN)-based composite models, and miscellaneous models that employ alternative architectures [19]. CNN-based composite models leverage different modules to extract features: utilizing CNN modules for capturing spatial dependencies, while employing sequential modeling components such as Recurrent Neural Networks (RNNs) to model temporal correlations. Yan et al. [20] developed a CNN-LSTM hybrid model to address the challenges of multi-time, multi-site air quality forecasting. Elbaz et al. [21] introduced a deep learning architecture that integrates three-dimensional CNN with Gated Recurrent Unit (GRU) and attention mechanisms, demonstrating superior accuracy in multi-horizon predictions and enabling reliable early warning systems for air pollutant events.

Similarly, GNN-based composite models separate the extraction of temporal and spatial features. They leverage GNN to capture spatial dependencies, as GNNs excel at processing graph-structured data and modeling complex relationships between monitoring

stations as nodes in a graph, while incorporating temporal modeling components to analyze the evolution of pollution patterns over time. Qi et al. [22] developed a hybrid model combining GNN and LSTM to effectively predict PM_{2.5} concentrations using spatio-temporal air quality data. Ding and Noh [23] proposed a hybrid model based on interpretable neural networks and GNN model for multi-step air quality prediction. Liu et al. [24] proposed a self-tuning spatio-temporal neural network architecture for air quality prediction, which consists of parallel modules: GNN for spatial dependency modeling, GRU for temporal feature extraction, coupled with a gating mechanism for adaptive feature fusion and a hyperband optimization module for automated hyperparameter tuning.

The miscellaneous models, which do not strictly adhere to CNN or GNN architectures, often integrate multiple techniques or propose novel architectures to address specific challenges in air quality modeling. Several studies have explored innovative combinations of traditional methods with deep learning approaches. Xu et al. [25] developed a novel deep learning method which combines a multi-source Gaussian diffusion model with a GRU network to predict atmospheric environmental quality data, particularly PM_{2.5} concentrations. These composite models, such as CNN-based, GNN-based, and hybrid approaches, use separate modules to learn spatial and temporal features. However, they encounter a significant challenge: their higher-level layers find it difficult to distinguish between temporal and spatial features and integrate them effectively across various scales and complexities [26]. To address this challenge, researchers have proposed several innovative models for spatio-temporal prediction. Wan et al. [26] introduced the correlated time series-oriented LSTM, an advanced recurrent network designed to collectively predict future values across spatio-temporal time series. However, this architecture with its multiple gates not only significantly increases computational costs but also constrains flexibility of model, making it less adaptable to varying temporal patterns, spatial distributions, and dynamic environmental conditions.

In response to the previously discussed limitations, Transformer-based models have emerged as particularly promising alternatives. The Transformer architecture [27] has demonstrated superior performance across multiple domains, including natural language processing, computer vision, speech, and time series analysis. Its attention mechanism, which automatically recognizes correlations between elements in a sequence, makes it particularly effective for sequential modeling problems [28]. The evolution of Transformer architectures for time series forecasting has followed two main research directions. The first focuses on adapting core components and architecture to better suit time series tasks. In this direction, Zhou et al. [29] proposed the Informer model which can address fundamental challenges including quadratic time complexity, high memory usage, and encoder-decoder architectural limitations. Wu et al. [30] proposed the Autoformer model which integrate a decomposition architecture for temporal pattern identification and introduce an efficient Auto-Correlation mechanism for dependency detection. The second research direction aims to fully leverage Transformer capabilities by addressing the inherent characteristics of time series data. This stream has produced several key innovations: Kim et al. [31] developed a flexible and efficient reversible instance normalization layer to address distribution shift issues, while Liu et al. [32] introduced Non-stationary Transformers, incorporating Series Stationarization and

De-stationary Attention to significantly enhance performance. Han et al. [33] further advanced this direction by exploring channel-dependent and channel-independent strategies, ultimately developing an enhanced Predict Residuals with Regularization method. Taking a novel approach, Liu et al. [34] demonstrated that substantial improvements could be achieved by adopting components on inverted dimensions with an altered architecture, suggesting that innovative structural modifications can unlock new performance potential.

Inspired by advances in Transformer architecture, researchers have progressively adapted and refined this framework to address the unique challenges of spatio-temporal air pollutant prediction. To handle the complexity of covering thousands of locations with fine spatial granularity, Liang et al. [35] introduced AirFormer, an innovative Transformer-based model for nationwide air quality prediction in China. This model employs a two-stage learning process: a bottom-up deterministic stage with novel self-attention mechanisms for efficient spatio-temporal representation learning, and a top-down stochastic stage using latent variables to capture data uncertainty. Building upon the Informer framework, Feng et al. [18] developed an enhanced spatio-temporal Informer model that incorporates novel spatio-temporal embedding and attention mechanisms, demonstrating improved performance in capturing complex spatio-temporal correlations across multiple monitoring stations. Yu et al. [36] further advanced this line of research by proposing a multi-granularity spatio-temporal fusion Transformer, which integrates three key components: a residual de-redundant block to eliminate information redundancy, a spatio-temporal attention block to capture inter-station correlations, and a dynamic fusion block for integrating multi-granularity predictions.

However, these approaches, while increasingly sophisticated, tend to design specialized modules for specific scenarios, resulting in increasingly complex model architectures. Consequently, they face significant challenges in balancing the trade-off between model complexity and prediction accuracy. This challenge is further compounded by the intricate nature of air pollution dynamics, where present and future pollutant concentrations are not only dependent on historical values but are also intricately linked to a complex web of meteorological conditions and spatial factors [14, 37]. These multifaceted dependencies, coupled with the spatial diversity and correlation patterns across monitoring stations, underscore the need for more elegant modeling approaches that can better capture these complex relationships without resorting to overly complicated architectures.

Moreover, air pollutants often have a non-stationary nature, particularly at finer temporal resolutions, and exhibit complex spatio-temporal relationships [14, 38, 39]. For instance, imagine that Station A is situated on the west side of the city, while Station B is on the east side. When the wind blows from west to east, common air pollutants like PM_{2.5} will show up at Station B later than at Station A. Specifically, as the wind carries the pollutants from west to east, the levels of PM_{2.5} at Station A will increase or decrease first, followed by similar changes at Station B after a certain period. This phenomenon emphasizes that air pollutant concentrations exhibit spatial correlations and temporal lead-lag relationships. In addition, the influence of meteorological factors on air pollution cannot be overstated. Variables such as wind speed and temperature play crucial roles in the dispersion, formation, and transformation of air pollutants. For

example, higher temperatures may accelerate the formation of ground-level ozone, while strong winds can disperse particulate matter more rapidly. Therefore, it is crucial to fully consider the spatio-temporal correlations between monitoring stations, the non-stationary characteristics of pollutants, and the impact of meteorological factors to build an effective air pollutant prediction model.

In traditional Transformer neural networks, tokens are assigned positional embedding, typically calculated using sine or cosine functions [27]. The Time2Vec embedding method is proposed for time series tasks to capture temporal characteristics. It encodes time using calendar components like year, month, day, season, hour, and minute [40]. [41] and [16] use Time2Vec embedding to predict PM_{2.5} indices in spatio-temporal sequences. Additionally, some studies, such as [42], combine position embedding and Time2Vec embedding for spatio-temporal predictions. However, the tokenization and data embedding steps do not fully utilize background spatial information. Moreover, unifying different variables into time-based tokens can affect air pollutants prediction accuracy because lead-lag relationships exist between the sequences.

To address the aforementioned problem, this study proposes a novel Spatio-Temporal inverted Transformer (ST-iTransformer). The proposed model flexibly captures spatial and temporal dynamics while fully accounting for the influence of meteorological conditions. “Inverted” explicitly reflects the approach of embedding the entire time series of each variable independently into a separate token, rather than embedding multiple variables at each timestamp into a single temporal token. This inversion strategy aggregates global representations for each variable, enabling a variate-centric approach that effectively utilizes attention mechanisms for capturing multivariate correlations [34]. Overall, the proposed model makes the following four significant contributions.

Firstly, the ST-iTransformer model is designed to address the challenges of forecasting air pollutants using spatio-temporal time series data. This model integrates the inherent non-stationary time dependencies and spatial information via the end-to-end way, providing a comprehensive approach to the complex task of air pollution prediction.

Secondly, a novel implementation of spatio-temporal embedding is proposed. The embedding approach addresses two main drawbacks of current Transformer-based models: the neglect of spatial background information and the use of absolute temporal position embedding. To address the first drawback, the geographic locations of monitoring stations, including longitude and latitude, are considered and embedded into the latent space. To address the second drawback, each time series is independently embedded into variate tokens, enabling the model to learn beyond uniform timestamps across different stations.

Thirdly, this research employs attention mechanisms and feed-forward network strategies on inverted dimensions. Following the embedding layer, the attention mechanism can identify correlations across multiple variables at different monitoring stations. Additionally, a feed-forward network processes each variate token to learn nonlinear representations. To the best of our knowledge, this is the first time the inverted Transformer framework has been utilized for air pollutant concentrations prediction.

Finally, the effectiveness of the proposed ST-iTransformer model is evaluated through comparative experiments, ablation studies, and analyses of different forecasting steps.

The evaluations are systematically conducted in Beijing, China. The experimental results demonstrate that the proposed model outperforms the benchmark models.

Methodology

Objective of ST-iTransformer

The ST-iTransformer model addresses the challenging task of spatio-temporal air quality forecasting. The data is collected from m monitoring stations, where each station captures l environmental variables, comprising s air quality indicators and p meteorological variables. The observational data structure follows a hierarchical organization. At the fundamental level, measurements at time step t are encoded in matrix form $\mathbf{X}_t \in \mathbb{R}^{l \times m}$, where rows index the measured variables and columns represent monitoring stations. The temporal sequence of these observations naturally forms a three-dimensional tensor $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n) \in \mathbb{R}^{n \times l \times m}$, integrating the temporal, variable, and spatial dimensions of the monitoring data. Given the historical observations denoted as $\mathbf{X}$, the model generates predictions for k future time steps, formally structured as $\mathbf{X}_{\text{output}} = (\mathbf{X}_{n+1}, \mathbf{X}_{n+2}, \dots, \mathbf{X}_{n+k}) \in \mathbb{R}^{k \times s \times m}$. Notably, while the model processes both air quality and meteorological data as input, its forecasting output specifically focuses on air quality variables, reflecting its specialized purpose in air quality prediction.

To facilitate efficient computational processing and enhance the extraction of spatio-temporal features, a dimensional transformation is applied to restructure the three-dimensional tensor into a two-dimensional matrix representation. This reshaping operation arranges the data by placing each variable from each station into a single column, with time steps as rows. Through this transformation, the core objective can be formulated as: given the historical observations $\mathbf{X} \in \mathbb{R}^{n \times lm}$, the model aims to generate predictions $\mathbf{X}_{\text{output}} \in \mathbb{R}^{k \times sm}$. This dimensional reduction strategy not only streamlines the computational process but also preserves the essential spatio-temporal relationships in the data. Moreover, the transformed representation enables more effective feature extraction and pattern recognition across multiple monitoring stations and time periods.

Overview of ST-iTransformer

The overall process of the ST-iTransformer model is depicted in Fig. 1 and Algorithm 1. Specifically, the ST-iTransformer model takes air pollutants and meteorological variables from multiple monitoring stations as inputs. The model begins by stabilizing the input through non-stationary normalization to address distribution shifts. Subsequently, the data is processed through an embedding layer, which comprises two components: one that embeds the entire series as a token and another that embeds geographical information. Then, the model employs the encoder architecture. The encoder comprises multiple blocks, each integrating attention mechanisms, feed-forward layers, and layer normalization. Finally, the encoded representations pass through a linear transformation layer, followed by a de-normalization process that restores the predictions to their original scale.

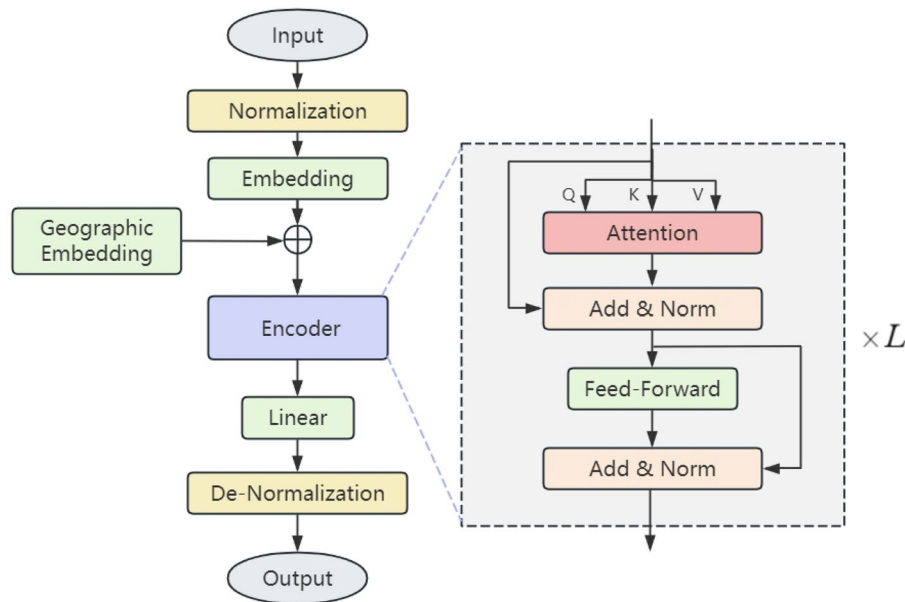


Fig. 1 Overview of ST-iTransformer

Algorithm 1 ST-iTransformer—overall architecture

Input: Input lookback time series $\mathbf{X} \in \mathbb{R}^{n \times lm}$; input Length n ; predicted length k ; observation stations m ; variables number at each station l ; token dimension D ; iTransformer block number L .

```

1:  $\mathbf{X} = \text{Normalization}(\mathbf{X})$ 
2:  $\mathbf{X} = \mathbf{X}.\text{transpose}$ 
3: Embed series into variate tokens:  $\mathbf{H}_T = \text{Linear}(\mathbf{X})$ 
4: Construct the geographical matrix:  $\mathbf{L}_s \in \mathbb{R}^{lm \times 2}$ 
5: Embed the geographical matrix:  $\mathbf{L}'_s = \text{Linear}(\mathbf{L}_s)$ 
6: Combine the two embedding parts:  $\mathbf{H}^0 = \mathbf{H}_T + \mathbf{L}'_s$ 
7: Encoder:
8: for  $i$  in  $\{1, \dots, L\}$  do
9:    $\mathbf{H}^{i-1} = \text{LayerNorm}(\mathbf{H}^{i-1} + \text{MultiHead}(\mathbf{H}^{i-1}))$ 
10:   $\mathbf{H}^i = \text{LayerNorm}(\mathbf{H}^{i-1} + \text{FFN}(\mathbf{H}^{i-1}))$ 
11: end for
12: Project tokens back to predicted series:  $\mathbf{X}'_{\text{output}} = \text{Linear}(\mathbf{H}^L)$ 
13:  $\mathbf{X}'_{\text{output}} = \mathbf{X}'_{\text{output}}.\text{transpose}$ 
14:  $\mathbf{X}_{\text{output}} = \text{De-normalization}(\mathbf{X}'_{\text{output}})$ 
15: return  $\mathbf{X}_{\text{output}}$ 

```

The modules of ST-iTransformer

Embedding

The embedding layer, as shown in Fig. 2, comprises two key components. The first component embeds the entire time series as a token. It captures holistic temporal patterns of inputs. The second component adds key geographical information, such as longitude

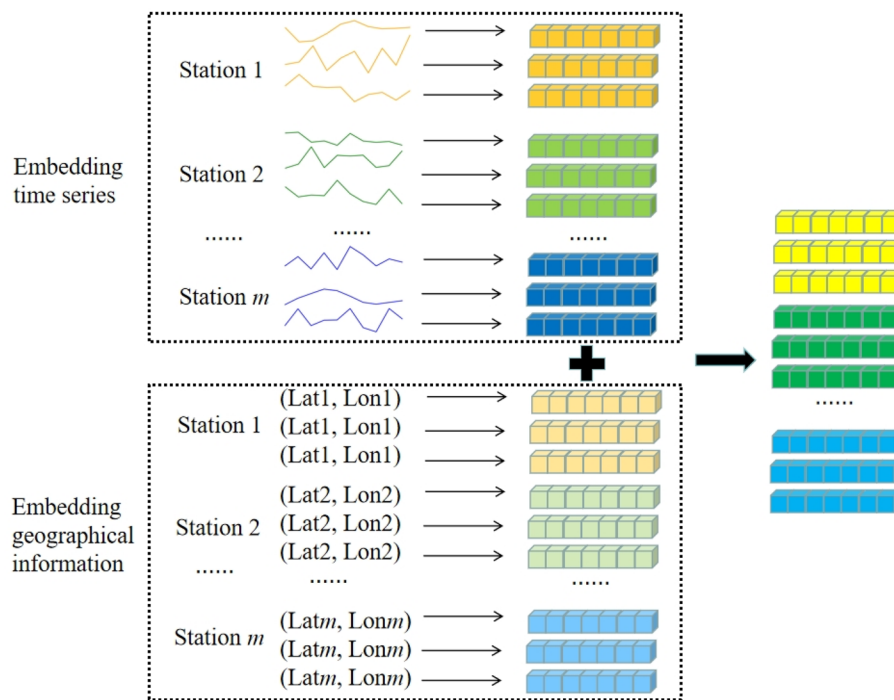


Fig. 2 Illustration of the embedding layer process

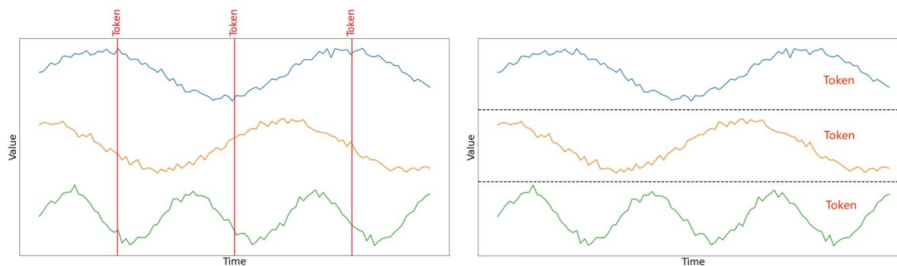


Fig. 3 The vanilla Transformer Embedding (left) and the inverted Transformer Embedding (right)

and latitude, providing crucial spatial context for understanding the dispersion of variables across different locations. These two components are combined through addition.

The first component projects each time series into a D -dimensional space. This design avoids embedding information from different variables at the same time point into a single token, which prevents the model from mistaking simultaneous occurrences for true temporal dependencies. This is particularly important when learning the lead-lag relationships between air pollutants across different monitoring stations. From an intuitive perspective, the entire time series of each variable at each monitoring station is independently embedded into a distinct token (right side of Fig. 3), instead of combining the multivariable data of a single time point into a token (left side of Fig. 3). Mathematically, the process begins by transposing the input $\mathbf{X} \in \mathbb{R}^{n \times lm}$, resulting in $\mathbf{X}^T \in \mathbb{R}^{lm \times n}$. This transposed input is then passed through a linear layer, given by $\mathbf{H}_T = \mathbf{X}^T \mathbf{W}_e^T + \mathbf{b}_e$, where $\mathbf{W}_e \in \mathbb{R}^{D \times n}$ is a learned weight matrix and $\mathbf{b}_e \in \mathbb{R}^D$ is a bias vector. These parameters are optimized during model training. This design enhances the embedded tokens by focusing on the variables, enabling

the attention mechanism to capture correlations among spatial multivariable data more effectively [34].

The second component focuses on fully leveraging the spatial correlations. It maps the longitude and latitude of all stations into a D -dimensional space. Specifically, a geographical matrix $\mathbf{L}_s \in \mathbb{R}^{lm \times 2}$ is constructed, and through a Linear projection, a latent feature matrix $\mathbf{L}'_s \in \mathbb{R}^{lm \times D}$ is obtained. This additional spatial context enhances the ability of the model to understand and predict air quality patterns across different locations.

Finally, comprehensive representations $\mathbf{H} = \mathbf{H}_T + \mathbf{L}'_s = \{\mathbf{h}_1, \dots, \mathbf{h}_m\} \in \mathbb{R}^{lm \times D}$ are obtained by combining the two components. This creates a multidimensional representation, which serves as a comprehensive input for the subsequent model layers, enhancing both the learning and prediction phases with greater accuracy.

Attention mechanism

The attention mechanism is utilized to identify complex relationships among various air pollutant variables from different monitoring stations. Specifically, linear projections are applied to generate queries, keys, and values: $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{l \times d_k}$, where d_k is the dimensionality of the projection. The scaled dot-product attention is given by,

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \quad (1)$$

Moreover, multi-head attention equips the model to simultaneously focus on different aspects of the input. Rather than depending on a single attention function, the ST-iTransformer employs h distinct sets of learned projections, enabling it to capture a wider range of input features. Formally, the multi-head attention mechanism can be expressed as,

$$\text{MultiHead}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)\mathbf{W}^O \quad (2)$$

$$\text{head}_i = \text{Attention}(\mathbf{Q}^{(i)}, \mathbf{K}^{(i)}, \mathbf{V}^{(i)}) \quad (3)$$

where $\mathbf{Q}^{(i)}, \mathbf{K}^{(i)}, \mathbf{V}^{(i)} \in \mathbb{R}^{l \times d_k}$ are h sets of distinct query, key, and value matrices, and $\mathbf{W}_i^O \in \mathbb{R}^{hd_k \times d}$ is the parameter matrix.

Encoder

Each encoder block in the ST-iTransformer architecture is composed of two sub-layers. The first layer is a multi-head attention mechanism, and the second layer is a simple position-wise fully connected feed-forward network (FFN). A residual connection is applied around each sub-layer, followed by layer normalization. This process can be formally expressed as,

$$\mathbf{H}^{i-1} = \text{LayerNorm}(\text{MultiHead}(\mathbf{H}^{i-1}) + \mathbf{H}^{i-1}) \quad (4)$$

$$\mathbf{H}^i = \text{LayerNorm}(\text{FFN}(\mathbf{H}^{i-1}) + \mathbf{H}^{i-1}) \quad (5)$$

where $\mathbf{H}^{i-1} \in \mathbb{R}^{lm \times D}$ represents the input to the l -th layer, and $\mathbf{H}^i \in \mathbb{R}^{lm \times D}$ denotes its output. MultiHead(.) denotes the multi-head attention mechanism, which aims to

capture dependencies between variates. LayerNorm(.) represents Layer Normalization [43], which is specifically adopted on series representations to mitigate discrepancies between variates. FFN(.) is the feed-forward network applied to the series representation of each variate token. The FFN adds non-linearity and complexity to the model, enabling it to learn temporal relationships. Additionally, it helps transform the input data into a more useful form for subsequent layers in the network. This iterative process is repeated for i in $\{1, \dots, L\}$, where L is the total number of encoder blocks, allowing the model to capture increasingly complex dependencies and representations within the input.

Non-stationary normalization and de-normalization

Since air quality time series often exhibit the distribution-varying problem accompanied by non-stationarity, which makes forecasting even more intractable, stationarization methods are crucial as pre-processing steps for deep model inputs. Based on this consideration, the proposed model adopts Non-stationary normalization and de-normalization methods, as introduced by [32]. This approach consists of a Normalization module using a sliding window over time and a De-normalization module transforming model outputs to final forecasting results.

The Normalization module, positioned at the beginning of the model. It is conducted on the temporal dimension using a sliding window. For each input series $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n) \in \mathbb{R}^{n \times lm}$, the normalization process is given by,

$$\mathbf{X}'_i = \frac{1}{\sigma_X} \odot (\mathbf{X}_i - \mu_X), \quad i = 1, 2, \dots, n \quad (6)$$

where $\mu_X = \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i$, $\sigma_X^2 = \frac{1}{n} \sum_{i=1}^n (\mathbf{X}_i - \mu_X)^2$, and $\odot$ indicates element-wise multiplication. Then, the result $\mathbf{X}' = (\mathbf{X}'_1, \mathbf{X}'_2, \dots, \mathbf{X}'_n) \in \mathbb{R}^{n \times lm}$ is obtained.

At the end of the model, the De-normalization module converts the processed data back to its original scale, ensuring the output predictions of air pollutant levels are accurately represented. It transforms the model output $\mathbf{X}'_{\text{output}} = (\mathbf{X}'_{n+1}, \mathbf{X}'_{n+2}, \dots, \mathbf{X}'_{n+k}) \in \mathbb{R}^{k \times lm}$ using μ_X and σ_X . The De-normalization process is given by,

$$\mathbf{X}_j = \sigma_X \odot \mathbf{X}'_j + \mu_X, \quad j = n+1, n+2, \dots, n+k. \quad (7)$$

Then, the forecasting results $\mathbf{X}_{\text{output}} = (\mathbf{X}_{n+1}, \mathbf{X}_{n+2}, \dots, \mathbf{X}_{n+k}) \in \mathbb{R}^{k \times lm}$ is obtained.

The data analysis

Research area and data descriptions

Beijing, the capital city of China, is situated between 39°54'N to 40°14'N latitude and 116°14'E to 116°34'E longitude, on the northern edge of the North China Plain. It is surrounded by the Yanshan Mountain range to its west, north, and northeast. Over recent decades, Beijing has experienced rapid industrialization, urbanization, and increased motorization [44]. This growth has resulted in rising energy consumption and increased emissions of multiple pollutants, significantly impacting air quality, public health, and the ecological environment. Particulate matter pollution has emerged as the most critical air quality issue in Beijing [45].

The study used data from 12 air quality monitoring stations in Beijing. The locations of these stations, represented by red circles, are shown in Fig. 4. Each station includes

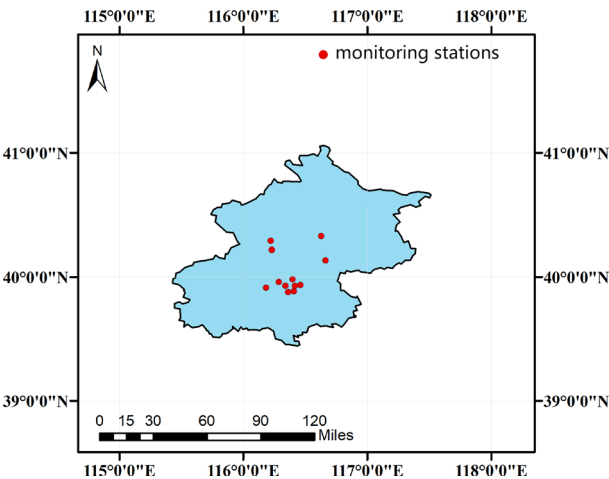


Fig. 4 Distribution of air pollution monitoring stations in Beijing

Table 1 The description of Dataset

Dataset	Beijing multi-site air quality dataset
Station number	12
Station name	Aotizhongxin, Changping, Dingling, Dongsi, Guanyuan, Gucheng, Huairou, Nongzhanguan, Shunyi, Tiantan, Wanliu, Wanshouxigong
Variable number	9 (5 pollutant variables and 4 meteorological variables)
Pollutant variables	PM2.5, PM10, SO2, NO2, O3
Meteorological variables	Temperature (TEMP), Dew Point (DEWP), Precipitation (RAIN), Wind Speed (WSPM)
Time range	2013/03/01 – 2017/02/28 (1461 days)
Frequency	Hourly
Dataset size	$1461 \times 24 \times 12 \times 9 = 3,786,912$

multiple pollutant variables, meteorological variables, and temporal information. Detailed information about the dataset, including specific monitoring station names, types of pollutants, and meteorological variables, is presented in Table 1. The data can be obtained from the website of the UC Irvine Machine Learning Repository (<https://archive.ics.uci.edu/dataset/501/beijingmultisiteairqualitydata>). The summary of average values for variables collected from 12 monitoring stations is presented in Table 2.

Data analysis

To gain a more thorough understanding of the spatio-temporal variations in air pollutant data and the influence of meteorological factors on pollution levels, multiple analyses are conducted.

In the spatial dimension, the global Moran’s *I* statistic for PM2.5 across multiple monitoring stations in Beijing is 0.2173 ($p < 0.0001$), indicating a significant spatial positive correlation and spatial clustering of air pollution. The global Moran’s *I* values for other pollutants also show different spatial correlation patterns. The Moran’s *I* value for PM10 is 0.1630 ($p < 0.005$), suggesting a moderate but significant spatial correlation. SO2

Table 2 The summary of average values for variables from 12 Monitoring Stations in Beijing

Type	Variable	Unit	Min	Max	Mean	Median	Std.
Pollutant variables	PM2.5	$\mu g/m^3$	2.0	999.0	79.84	55.0	80.95
	PM10	$\mu g/m^3$	2.0	999.0	104.91	82.0	92.43
	SO2	$\mu g/m^3$	0.28	500.0	15.91	7.0	21.89
	NO2	$\mu g/m^3$	1.03	290.0	50.60	43.0	31.79
	O3	$\mu g/m^3$	0.21	1071.0	57.24	44.0	57.14
Meteorological variables	TEMP	°C	− 19.9	41.6	13.53	14.5	11.44
	DEWP	°C	− 43.4	29.1	4.23	4.9	13.80
	RAIN	mm	0.0	72.5	0.06	0.0	0.82
	WSPM	m/s	0.0	13.2	1.73	1.4	1.25

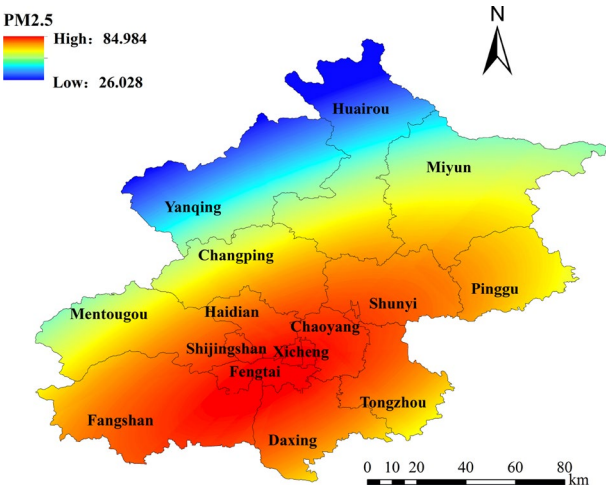


Fig. 5 Spatial Distribution of PM2.5 Concentration in Beijing

and NO2 both exhibit significant spatial clustering, with identical Moran’s *I* values of 0.1914 ($p < 0.005$). For O3, the spatial autocorrelation is minimal (Moran’s $I = 0.0082$, $p = 0.0571$).

The ordinary Kriging interpolation using the circular model is conducted for the average PM2.5 concentration, as shown in Fig. 5. The results indicate that PM2.5 levels are higher in the central and southern parts of Beijing. In contrast, the PM2.5 levels are lower in the northwestern districts. Furthermore, according to the local Moran’s *I* statistic, as shown in Fig. 6, multiple monitoring stations in the central and southern districts form high-high clusters, indicating severe pollution hotpots. Meanwhile, some stations in the northwest form low-low clusters, representing cleaner areas. The decreasing trend of PM2.5 concentration from southeast to northwest is likely attributable to the combined effects of population density, traffic emissions, and industrial activities concentrated in the southeastern districts. The results of the local Moran’s *I* statistic for other pollutants can be found in the supplementary material (see Fig. S1-S4) and exhibit patterns similar to those observed for PM2.5.

To further explore the dynamic relationships between the pollution indicators of different stations, four time periods of 100 time points, or 100 h, are randomly

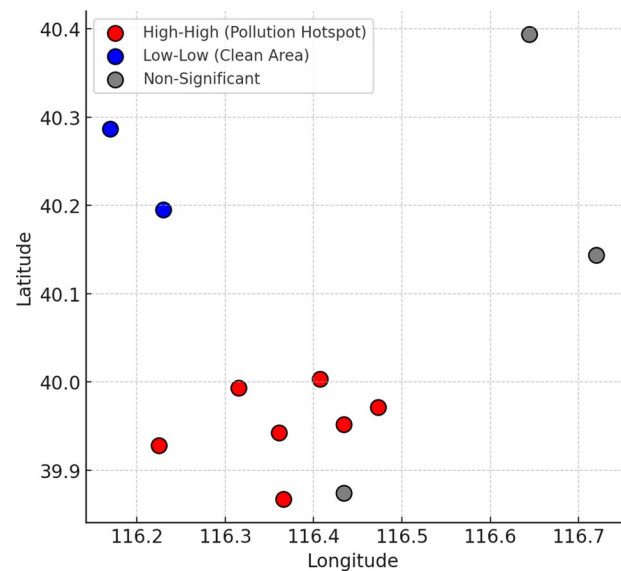


Fig. 6 Local Moran's I Clusters for PM2.5 Concentration in Beijing

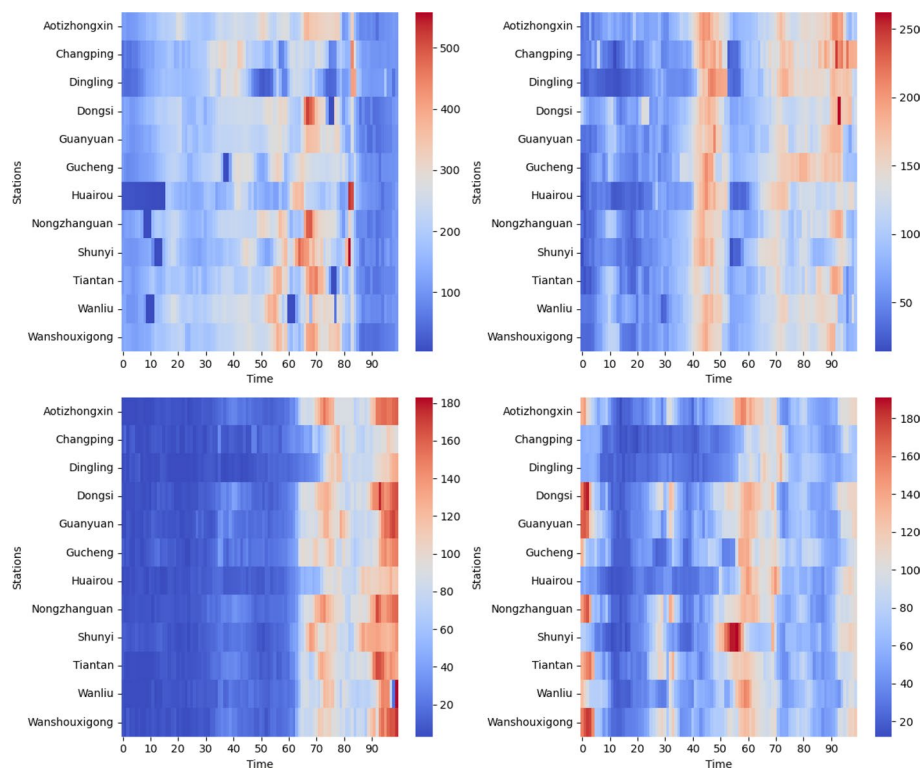


Fig. 7 Heatmap of 100-Hour periods across stations

selected to create heatmaps for PM2.5, as shown in Fig. 7. The horizontal axis of the heatmap represents time, and the vertical axis represents stations. The heatmap analysis of PM2.5 values across time and stations reveals complex lead-lag relationships in air pollution patterns of Beijing. High PM2.5 concentrations, represented by lighter

colors and red patches, shift between stations over time. This progression suggests that pollution events originate at specific stations and later affect others. Although many stations exhibit similar PM2.5 fluctuations, slight time differences point to the presence of lead-lag effects. These observations underline the dynamic spatio-temporal nature of PM2.5 pollution in Beijing. When variables are analyzed solely by time points, neglecting lead-lag relationships may compromise prediction accuracy. Therefore, a deeper understanding of these relationships is crucial for improving predictions of citywide air quality and tracing pollutant dispersion.

The correlation coefficients for air pollutants and meteorological variables across all stations are calculated and presented. Figure 8 illustrates the results for the Aotizhongxin station as a representative example, while corresponding results for the other stations can be found in the supplementary material (see Fig. S5–S15). As shown in Fig. 8, PM2.5 and PM10 exhibit a strong positive correlation (0.88), indicating they often increase together. PM2.5 also has a positive correlation with NO2 (0.68), while O3 exhibits a significant negative correlation with NO2 (−0.50) and SO2 (−0.21). Additionally, TEMP is positively correlated with O3 (0.59) and strongly correlated with DEWP (0.82). The correlation matrix also reveals a moderate negative correlation between NO2 and WSPM (−0.49), which suggest that higher wind speeds tend to reduce NO2 levels. The results of correlation coefficients for other stations similarly demonstrate interactions between pollutants and meteorological factors. Therefore, these interactions should be considered when constructing models for pollutant prediction.

Pre-processing process

Missing data is filled by averaging the values from the previous day and the following day. The choice of this interpolation method is based on the characteristics of the dataset

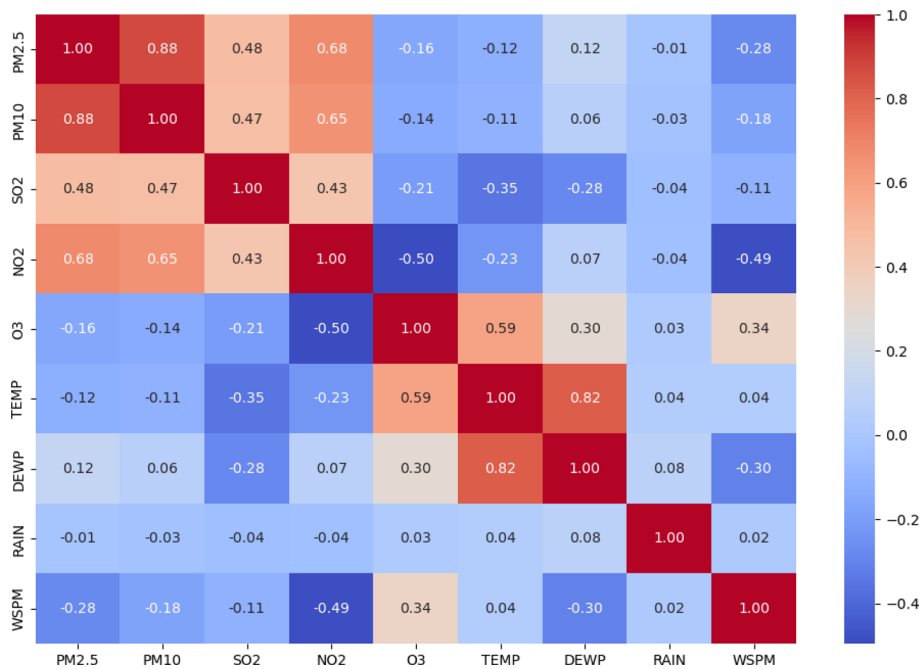


Fig. 8 Correlation coefficients for air pollutants and meteorological variables at Aotizhongxin

in this study. Specifically, the days with missing values account for less than 0.5% of the dataset, and these missing data are sporadically distributed throughout the entire dataset. The selected linear interpolation method is considered reasonable [46]. For outliers, they are retained since the data is deemed accurate and valid. On one hand, deep learning models offer a favorable balance between flexibility and effectiveness, and input data requirements are less strict [47]. On the other hand, outliers are intentionally retained to assess the model's capability in predicting extreme values. This practice serves as a precautionary measure against extreme air pollution.

The dataset is composed of a three-dimensional vector that includes time, data from multiple air quality monitoring stations, and multidimensional data for each station. Through a reshaping operation, the dimension related to different air quality monitoring stations is compressed, with the data for each variable from each station placed into a single column and time as the rows. This results in a two-dimensional dataset. This data integration technique enhances the ability to extract information from multiple monitoring stations by increasing dimensions and consolidating the input data into a two-dimensional feature vector. The compression operations, by making it easier to identify key patterns and relationships, benefit subsequent information extraction and prediction processes.

Experiment

Experimental settings and evaluation indicators

The proposed ST-iTransformer is developed with Python 3.10 and the deep learning library PyTorch 2.0.1. Model training and validation are carried out on the Nvidia A16 GPU platform. The experimental procedure framework is shown in Fig. 9. Hyperparameters are adjusted based on the results from this process. The model employs the early stopping method, halting training if performance does not improve after three consecutive epochs. Hyperparameter combinations are explored and optimized using grid search during the training phase. The learning rate is tested with the values {0.001, 0.0001, 0.0002, 0.0005, 0.00001}, and the batch size varies among {32, 64, 128, 256}. The

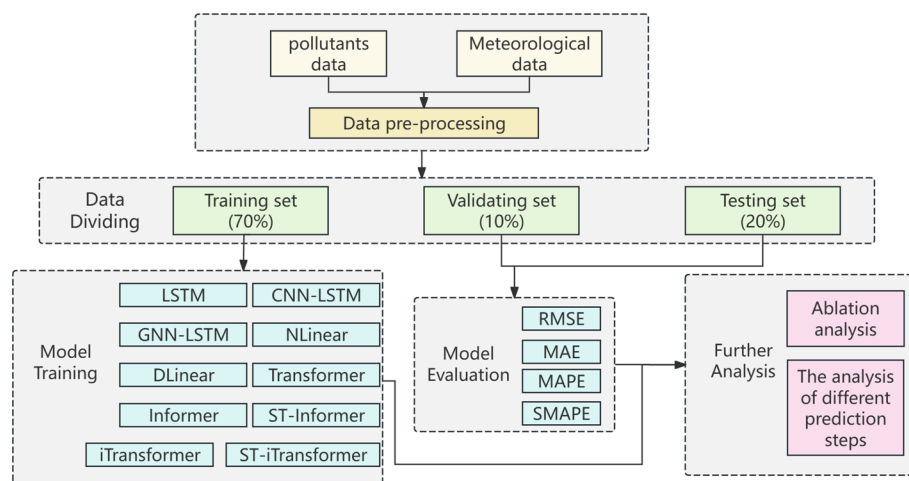


Fig. 9 The experimental procedure framework

number of encoder layers is explored with {1, 2, 3}. Additionally, the dimension of the series representations D is adjusted within the set {32, 64, 128, 256, 512}. The experimental training parameter settings are summarized in Table 3. Lookback lengths are set to 96, and prediction lengths are set to 24, analyzing past data over 96 time steps and predicting future data for the next 24 time steps. To ensure sufficient training data and evaluate prediction performance, the datasets are divided into a ratio of approximately 7:1:2 for the training, validation, and test sets.

Mean absolute percentage error (MAPE), mean absolute error (MAE), symmetric mean absolute percentage error (SMAPE), and root mean squared error (RMSE) are used to quantify the prediction performance of models. The equations for predicting k steps ahead are provided below:

$$\text{RMSE} = \sqrt{\frac{1}{k} \sum_{i=1}^k (y_i - \hat{y}_i)^2} \quad (8)$$

$$\text{MAE} = \frac{1}{k} \sum_{i=1}^k |y_i - \hat{y}_i| \quad (9)$$

$$\text{MAPE} = \frac{1}{k} \sum_{i=1}^k \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100\% \quad (10)$$

$$\text{SMAPE} = \frac{1}{k} \sum_{i=1}^k \frac{|y_i - \hat{y}_i|}{(|y_i| + |\hat{y}_i|)/2} \times 100\% \quad (11)$$

where y_i represents the observed value and $\hat{y}_i$ denotes the predicted value.

Baseline models

The proposed ST-iTransformer is evaluated against multiple state-of-the-art time series forecasting models to assess its forecasting performance. The baseline models are listed below,

Table 3 Experimental training parameters setting

Parameters	Value
Batch Size	64
Epochs	10
The dimension D	256
The number of encoder layers	3
Lookback lengths	96
Prediction lengths	24
Learning Rate	0.0002
Loss Function	MSE
Optimizer	Adam
Early Stopping	Patience: 3

- LSTM: The Long Short-Term Memory neural network, which is a special type of RNN model [48].
- CNN-LSTM: A spatio-temporal neural network that combines CNN for spatial feature extraction with LSTM for temporal sequence prediction [20].
- GNN-LSTM: A spatio-temporal neural network that integrates GNN for spatial feature extraction with LSTM for temporal sequence prediction [49].
- NLinear: A model that normalizes the input sequence by subtracting the last value, processes it through a linear layer, and then adds the subtracted part back for the final prediction [50].
- DLinear: A model that decomposes input data into trend and seasonal components, processes each with linear layers, and sums them up for the final prediction [50].
- Transformer: A neural network using self-attention to capture long-range dependencies in data [27].
- Informer: An efficient Transformer variant for long sequence time-series forecasting using a ProbSparse self-attention mechanism [29].
- ST-Informer: An Informer-based model that uses spatio-temporal attention to capture both spatial and temporal dependencies efficiently [18].
- iTransformer: An inversed transformer that applies the attention and feed-forward network on the inverted dimensions [34].

With the same parameter settings as the proposed model, the baseline models are configured with a lookback length of 96 and a prediction length of 24. The datasets are partitioned in a 7:1:2 ratio for training, validation, and testing. Regarding additional parameters, extensive grid search is conducted to tune the hyperparameters, with final configurations selected based on optimal performance metrics. The LSTM model employs a batch size of 64, a learning rate of 0.0002, a dropout rate of 0.3, a hidden size of 64, and 2 LSTM layers. In contrast, the CNN-LSTM model maintains the same batch size and dropout rate but adopts a higher learning rate of 0.0005, a hidden size of 128, and 3 LSTM layers, integrating a multi-scale CNN feature extraction module with kernel sizes of 3, 5, and 7. Similarly, the GNN-LSTM model utilizes the same batch size, dropout rate, hidden size, and learning rate as CNN-LSTM, with 2 LSTM layers, and 2 graph convolution layers, which leverage an adjacency matrix for structured feature extraction. The details are shown in the supplementary material (see Table S1). NLinear and DLinear both adopt a learning rate of 0.0002 to ensure stable convergence. DLinear further incorporates a decomposition kernel size of 25, enabling explicit separation of trend and seasonal components to enhance predictive performance. The Transformer-based models (Informer, ST-Informer, and iTransformer) follow a consistent parameter configuration, featuring a learning rate of 0.0002, a batch size of 64, a dropout rate of 0.3, and a dimension D of 256.

The performance analysis of ST-iTransformer

Table 4 presents a comparative analysis of the proposed method against various baseline models. Each experiment is conducted five times with different random seeds, and the mean and variance are reported. The proposed ST-iTransformer model consistently outperforms all other models across all metrics. Compared to the iTransformer,

the ST-iTransformer achieves a 2.57% reduction in RMSE and a 2.70% improvement in MAE, along with a 2.56% improvement in MAPE and a 5.11% reduction in SMAPE. These results highlight the effectiveness of the ST-iTransformer in capturing and modeling the complex spatio-temporal dynamics of air pollutant concentrations.

The paired t-test is performed on the error metrics, with the test results included in the supplementary material (see Table S2). The statistical analysis reveal that ST-iTransformer demonstrate statistically significant differences ($p < 0.05$) in the majority of comparisons with baseline models across all four evaluation metrics (RMSE, MAE, MAPE, and SMAPE). There are only a few exceptions: in RMSE comparisons, CNN-LSTM, GNN-LSTM, and ST-Informer have p -values slightly above 0.05, while in MAPE comparisons, iTransformer has p -values above 0.05. These results strongly indicate that the proposed model achieves statistically significant improvements over the baseline methods.

A comprehensive comparison of model performance in predicting five air pollutants across various monitoring stations is meticulously computed and presented in Fig. 10. The ST-iTransformer model demonstrates superior predictive capability, exhibiting lower error values for most pollutants and across the diverse array of monitoring stations. This superiority is quantitatively validated by the statistical analysis, which reveals that ST-iTransformer achieved the lowest RMSE in 53.30% of all evaluated scenarios and the lowest MAE in an impressive 66.70% of scenarios, substantially outperforming baseline models. These results indicate robust generalizability across both pollutant types and spatial domains. Additional error metric further corroborate these findings and is documented in the supplementary material (see Fig. S16).

Analysis also reveals that all models show significant fluctuations in error rates across different geographical locations, demonstrating that prediction accuracy exhibits notable spatial variability. To provide a comprehensive visualization of the spatial distribution of prediction errors, spatial interpolation maps of RMSE and MAE are generated using ordinary Kriging interpolation with the circular model, as shown in Fig. 11. The results indicate that higher prediction errors are concentrated in the central and southern parts of Beijing, which also exhibit elevated pollution levels. This pattern is likely attributed to the diverse emission sources and the complex dynamics of air pollution in these areas, leading to increased modeling uncertainty. In contrast, the northwestern

Table 4 Performance comparison of different models

Model	RMSE	MAE	MAPE	SMAPE
LSTM	50.15 $\pm$ 0.89	29.77 $\pm$ 0.16	167.96 $\pm$ 6.32	65.68 $\pm$ 0.48
CNN-LSTM	47.56 $\pm$ 1.19	27.78 $\pm$ 0.43	148.35 $\pm$ 2.42	60.30 $\pm$ 0.91
GNN-LSTM	46.99 $\pm$ 1.11	27.23 $\pm$ 0.49	143.70 $\pm$ 5.09	59.91 $\pm$ 0.66
NLinear	49.83 $\pm$ 0.05	29.09 $\pm$ 0.04	145.33 $\pm$ 0.23	57.38 $\pm$ 0.14
DLinear	48.31 $\pm$ 0.04	28.69 $\pm$ 0.02	162.34 $\pm$ 0.40	59.85 $\pm$ 0.09
Transformer	50.29 $\pm$ 1.30	29.28 $\pm$ 0.49	147.41 $\pm$ 5.69	67.04 $\pm$ 1.45
Informer	49.02 $\pm$ 1.10	28.55 $\pm$ 0.44	144.17 $\pm$ 6.62	65.00 $\pm$ 2.83
ST-Informer	46.38 $\pm$ 0.64	27.10 $\pm$ 0.42	137.27 $\pm$ 6.42	62.56 $\pm$ 3.86
iTransformer	47.45 $\pm$ 0.48	27.05 $\pm$ 0.30	121.36 $\pm$ 3.18	56.79 $\pm$ 1.56
ST-iTransformer	46.23 $\pm$ 0.22	26.32 $\pm$ 0.08	118.25 $\pm$ 1.31	53.89 $\pm$ 0.24

Bold indicates the best performance

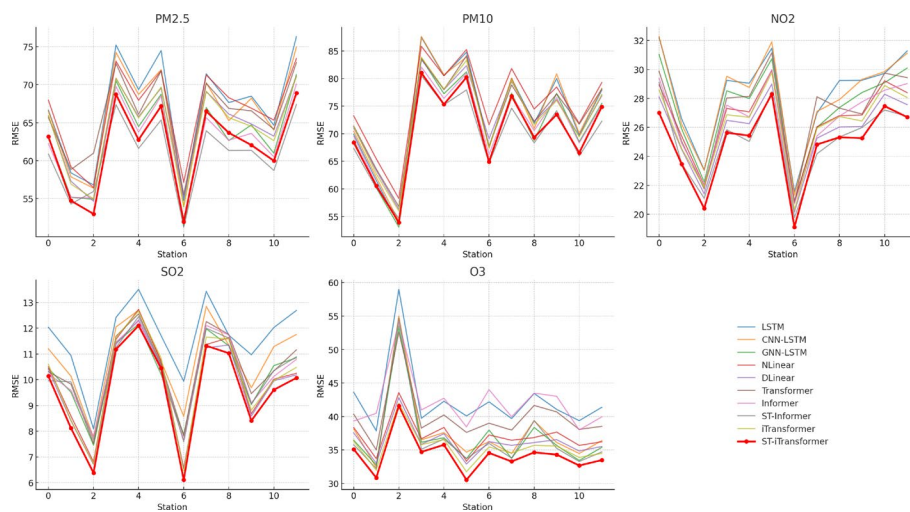


Fig. 10 Comparison of RMSE across stations for multiple pollutants using different models

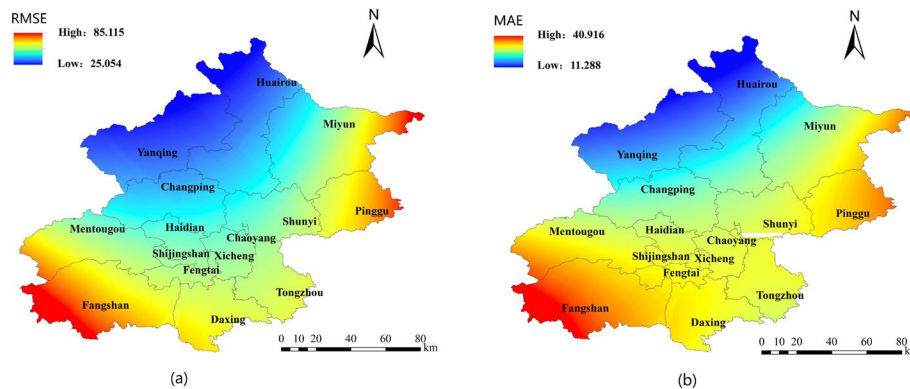


Fig. 11 Spatial distribution of prediction errors. **a** RMSE, **b** MAE

regions demonstrate relatively lower pollution levels, accompanied by smaller prediction errors. This may be due to more stable atmospheric conditions and a lower density of pollution sources, resulting in improved model performance.

Ablation study

Assessing the impact of the embedding layer

An ablation study is conducted to demonstrate the importance of two critical parts of the proposed embedding layer. The study tests different models as follows. The first is w/o GI, which is ST-iTransformer without embedding geographical information, specifically excluding longitude and latitude data. The second is w/o IV, which is ST-iTransformer without performing inversion. Ablation experiments use identical datasets and hyperparameters as the main study. Figure 12 presents min-max normalized results (range 1–10) of the ablation study, displaying the average RMSE and MAE of all pollutants, as well as the RMSE and MAE of each pollutant.

The ablation study shows that the base model ST-iTransformer, with the proposed embedding layer, achieves the best results across all metrics. Specifically, the model

w/o GI scores the lower accuracy. This demonstrates that embedding geographical information into the embedding layer is meaningful and helps the model learn spatial relationships more effectively. The accuracy of the model w/o IV is also noticeably lower than that of the ST-iTransformer. This demonstrates that the inversion technique enhances the ability of the model to learn the complex relationships between air pollutants across different stations. This finding aligns with the insights gained from heatmap Fig. 7, which revealed that air pollutants at different stations do not always exhibit synchronized variations and may display lead-lag relationships. Without employing the inverted perspective, embedding air pollutants data from different stations at the same time point would hinder the capacity of the model to effectively capture spatial relationships.

Further analysis indicates pollutant-specific dependencies: for PM_{2.5}, PM₁₀, and NO₂, removing geographical information leads to slightly higher prediction errors compared to removing the inversion technique. This suggests these pollutants have stronger spatial dependencies due to their strong association with local emissions. In contrast, for SO₂ and O₃, the absence of the inversion technique results in higher errors than removing geographical information, indicating these pollutants primarily rely on long-term temporal patterns and multivariate interactions effectively captured by the inversion technique. In summary, geographical information and inversion operations are two important parts of the embedding layer that contribute positively to our proposed model, with their relative importance varying across different pollutants.

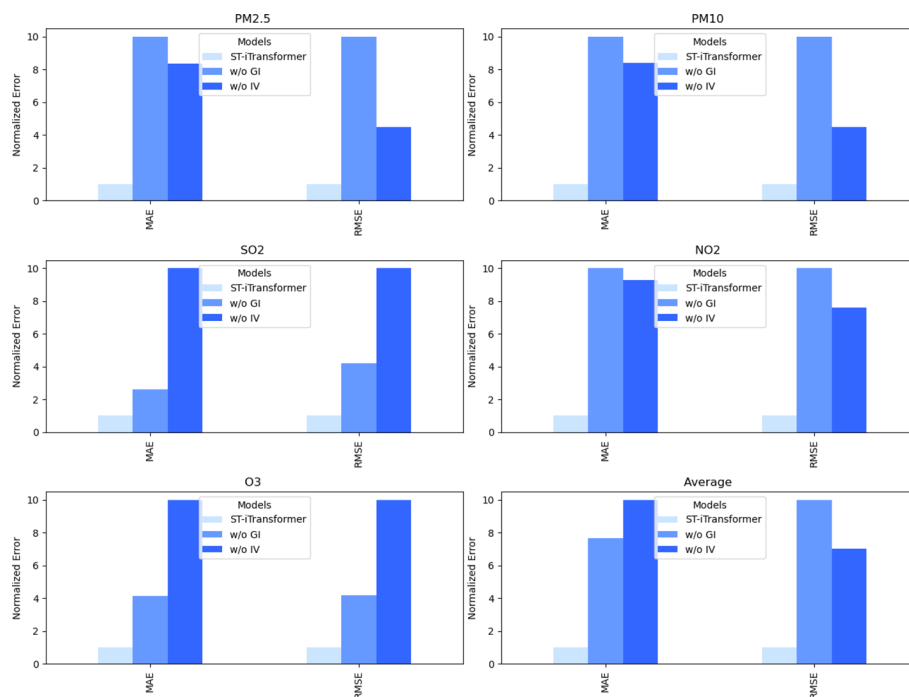


Fig. 12 Normalized errors of the ablation study of embedding layer

Examining the contribution of spatial data integration

The significance of incorporating pollutants and meteorological factors from other stations in predicting pollutants for target stations is evaluated. The experiments are divided into two groups: the first group uses only the data from the current station, conducting single-station (temporal) predictions, while the second group utilizes data from all 12 stations, conducting multi-station (spatio-temporal) predictions. The results, presented in Fig. 13 illustrate the comparative performance. Generally, lower error values are observed for the spatio-temporal predictions compared to the temporal predictions. The findings consistently demonstrate that spatio-temporal predictions outperform temporal predictions across various pollutants and stations. The superior performance of spatio-temporal model validates its efficacy in capturing complex pollution dynamics that may be influenced by regional factors and inter-station relationships. In conclusion, this ablation study provides compelling evidence for the necessity and effectiveness of spatio-temporal modeling in air pollutants prediction. Additional error metric further corroborate these findings and are comprehensively documented in the supplementary material (see Fig. S17).

Analyzing the influence of meteorological data

The ablation study further examines the impact of meteorological data on the performance of model. A variant of the ST-iTransformer, denoted as w/o MD, is evaluated by excluding meteorological variables such as TEMP, DEWP, RAIN, and WSPM. Results, as shown in Table 5 indicate that the model w/o MD exhibits inferior performance compared to the full ST-iTransformer. This observation underscores the importance of meteorological factors in pollutant concentration predictions. However, it is noteworthy that the improvement in accuracy, while positive, is not dramatic. On one hand, meteorological data often affects pollutant concentrations, so incorporating it helps improve prediction accuracy. On the other hand, the noise may accompany these exogenous variables. Therefore, the addition of meteorological factors does not significantly improve

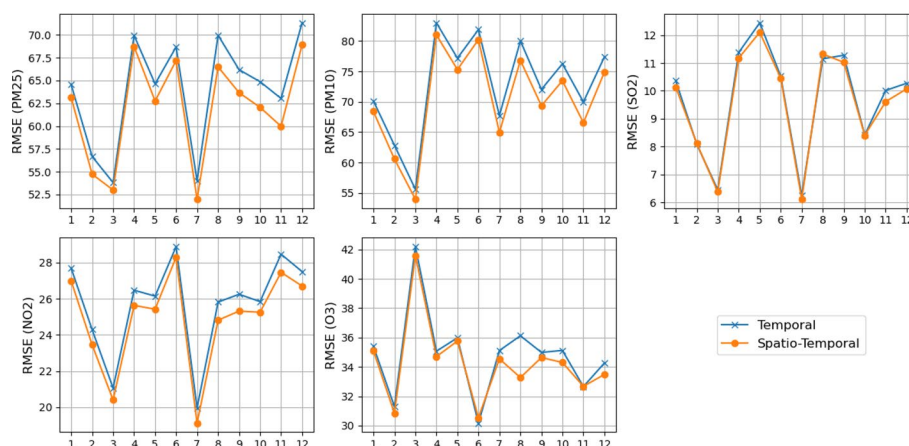


Fig. 13 Comparison of RMSE for temporal and spatio-temporal predictions

Table 5 Ablation study on meteorological data influence

Model	PM10		PM2.5		SO2		NO2		O3		Average	
	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE
w/o MD	63.59	40.89	72.23	47.48	9.97	4.95	25.88	18.61	35.21	24.69	47.47	27.33
Proposed	62.02	39.21	70.96	45.82	9.80	4.80	25.19	18.07	34.35	23.74	46.45	26.25

Bold indicates the best performance

accuracy. In conclusion, the inclusion of meteorological data, despite its potential for introducing noise, proves beneficial for improving prediction accuracy.

The analysis of different prediction steps

Precise prediction of air pollutant levels over extended periods is vital for safeguarding public health and informing daily activities. This complex forecasting process, involving multiple steps and timeframes, plays a crucial role in environmental management and urban planning. In multi-step forecasting, there are two main methods: iterated multi-step (IMS) forecasting and direct multi-step (DMS) forecasting. IMS forecasting uses a single-step model repeatedly when forecasting more than one time step ahead. DMS forecasting uses a single model optimized for forecasting multiple steps at once. Compared to IMS, DMS forecasting reduces error accumulation and speeds up model inference, making it better suited for long-term predictions [17]. Therefore, this study is based on DMS forecasting and analysis, which also helps provide early warnings for extreme pollution.

A large number of experiments are conducted to examine the performance of models across various prediction steps. With the lookback step fixed at 96, the prediction steps vary as 1, 2, 6, 12, 24, 48, 72, and 96, while other parameters remain consistent with the main experiment. Figure 14 illustrates the performance. The x-axis represents the number of prediction steps, while the y-axis shows the RMSE or MAE values. Notably, the proposed ST-iTransformer model demonstrates the lowest RMSE and MAE in most cases. This indicates that our proposed model outperforms the other models in the comparison in terms of prediction accuracy. Additionally, a notable trend is observed in the experimental results. As the prediction step length is increased, a corresponding increase in both the RMSE and MAE values is seen. This pattern is found to be consistent across all models tested, including the proposed ST-iTransformer. As forecasts are attempted further into the future, the uncertainty in predictions naturally accumulates. This increases the difficulty in accounting for potential changes in the underlying system dynamics. In summary, the ST-iTransformer model demonstrates better performance for both short-term and long-term forecasting, reflecting the robustness and stability of the model. However, all models still have inherent limitations in their ability to capture and project long-term trends and patterns.

Capability of predicting pollutants on unseen location

To evaluate the generalization capability of model in predicting the variables at unseen locations, experiments are conducted using data from varying numbers of stations (3, 6, 9, and all 12) to predict all 12 stations. As shown in Table 6, increasing the

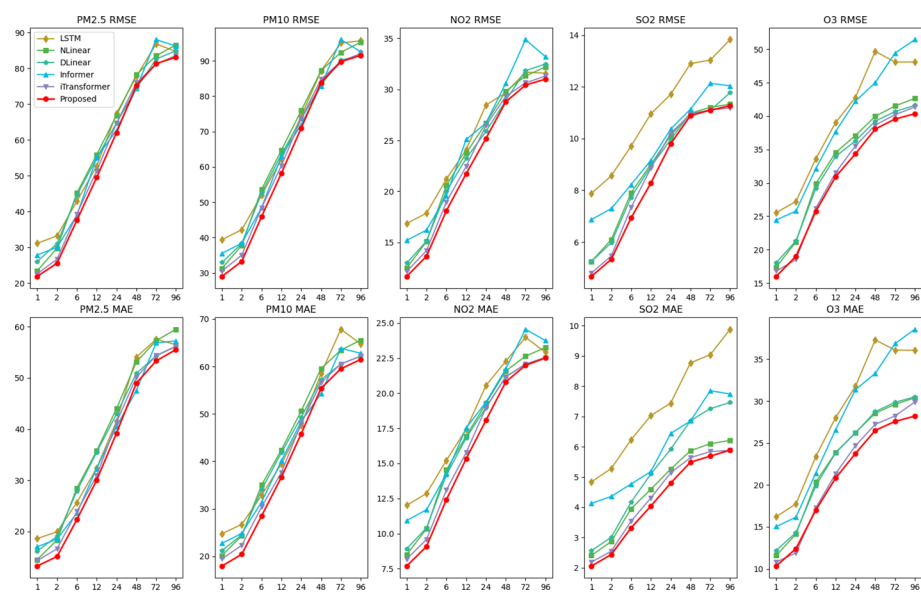


Fig. 14 Comparison of RMSE and MAE for different prediction steps

Table 6 Performance metrics with different numbers of training stations

Variables	RMSE	MAE	MAPE	SMAPE
27 (3 stations)	47.79	27.04	116.90	56.60
54 (6 stations)	47.41	26.91	114.62	55.46
81 (9 stations)	47.12	26.83	115.57	55.16
108 (all 12 stations)	46.46	26.33	115.81	53.73

Each station has 9 variables

number of training stations leads to a gradual reduction in error metrics. The performance improves as more station data are incorporated into the training process. However, the performance differences are relatively small, with only a 2.8% difference in RMSE between using 3 stations and all 12 stations. This minimal performance degradation when predicting unseen locations indicates that the proposed model effectively learns transferable time series representations that generalize well across locations. The ability to maintain robust performance even with limited training locations demonstrates its superior locations generalization capacity.

Discussion

The ST-iTransformer model performs well in forecasting air pollutants and consistently surpasses several state-of-the-art models. In station-level analyses, minor fluctuations in accuracy appear for specific pollutants at certain locations, but the ST-iTransformer generally maintains lower error rates across most stations. These results highlight its effectiveness in capturing and modeling the spatio-temporal relationships between air pollutant concentrations across diverse stations.

Ablation studies further emphasize the importance of key components in the ST-iTransformer. Including geographical information, such as longitude and latitude, proves

crucial for improving the accuracy of the model, as excluding it results in a noticeable drop in performance. The inversion technique, which captures lead-lag relationships between pollutants at different stations, also enhances the accuracy of the model. Additionally, the analysis of spatio-temporal versus temporal data reveals that predictions based on multi-station data consistently outperform those using single-station data, underscoring the importance of considering regional factors and inter-station relationships in pollutant forecasting. The inclusion of meteorological data, while beneficial, has a more moderate impact on prediction accuracy. Variables such as TEMP, DEWP, RAIN, and WSPM improve the performance of the model, but the gains are less significant, likely due to the noise introduced by these external factors.

Experiments on prediction length show that as the forecast period increases, both RMSE and MAE rise, which is typical of time series models due to accumulating uncertainties over time. Despite this, the ST-iTransformer maintains superior performance in both short-term and long-term predictions, demonstrating its overall robustness and ability to adapt to various forecasting challenges. The proposed model also demonstrates strong generalization capability across locations, maintaining robust performance with minimal error increase even when trained on limited station data.

Conclusion

The pollutants from different monitoring stations exhibit synchronized patterns for the majority of the time, while at other times, they display lead-lag relationships. To enhance the accuracy of predicting ambient air pollutants, the ST-iTransformer is proposed as a spatio-temporal multi-step prediction model. It employs an innovative embedding method consisting of two parts: the first involves embedding the entire time series, and the second involves embedding the longitude and latitude. Compared to other Transformer-based models, this research utilizes attention mechanisms and feed-forward network strategies on inverted dimensions. This approach addresses the problem of embedding different variables at the same time while ignoring the lead-lag relationships. Additionally, it cleverly incorporates the characteristics of spatial station data. ST-iTransformer excels in air pollutant forecasting across multiple stations and surpasses other state-of-the-art models. The key components of the model include the integration of geographical data, an inversion technique, and the use of multi-station data, while meteorological factors offer modest improvements. ST-iTransformer demonstrates robust performance in both short-term and long-term predictions and maintains lower error rates even as forecast periods extend, which highlights its adaptability to various forecasting scenarios.

The ST-iTransformer considers only certain meteorological conditions as exogenous variables, without accounting for other factors related to air pollutants, such as traffic, industrial activities, terrain barriers, topography and wind direction. Additionally, it does not integrate source apportionment data, which provides valuable insights into the contributions of different pollution sources. Future research could incorporate more types of data to enhance the ability of the model further. When designing the model, incorporating error-resilient mechanisms for sensor drift or localized weather anomalies could be beneficial, and more comprehensive approaches are worth exploring. Furthermore, the ST-iTransformer is designed and applied for air pollutant prediction in

this study. However, its versatility extends beyond air pollution forecasting. Its ability to capture complex spatial relationships makes it a valuable tool for researchers in various domains. Therefore, the ST-iTransformer may contribute to advancements in diverse areas where understanding and predicting spatially distributed phenomena are crucial.

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s40537-025-01150-5>.

Supplementary file 1.

Author contributions

All authors contributed to the study conception and design. Rui Zhang was responsible for conceptualization, methodology, formal analysis, and writing the original draft. Norhashidah Awang provided validation, supervision, and writing - review and editing. All authors read and approved the final manuscript.

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Data availability

The data can be obtained from the website of the UC Irvine Machine Learning Repository. (<https://archive.ics.uci.edu/dataset/501/beijingmultisiteairqualitydata>) Materials and methodologies are described in this paper.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

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